

Linear Inequalities

Clean quiz-style study book with graphs, mathematical formatting, step-by-step solutions, and SAT-calibrated difficulty.

How this version is built

Questions and answer choices are retyped cleanly.

Graphs are embedded where appropriate from the uploaded source document.

Fractions and division are shown in mathematical style instead of calculator-style slashes.

Each question is kept to one page when possible.

Difficulty rubric

Easy: direct translation, one-step range, simple proportional reasoning, or quick substitution.

Medium: systems, table checks, graph-solution checks, or whole-number constraints.

Hard: graph constant-finding, layered setup, piecewise or overtime logic, or careful constraint analysis.

1

Linear Inequalities - Regions in the Coordinate Plane

In the xy -plane, which of the following does NOT contain any points (x, y) that are solutions to $7x + 4y > 12$?

A) $x > 0$ and $y > 0$	C) $x < 0$ and $y < 0$
B) $x < 0$ and $y > 0$	D) $x > 0$ and $y < 0$

Answer: C

Short Explanation

When both x and y are negative, $7x + 4y$ is always negative, so it can never exceed 12.

Mathematical Shortcut

Plug in any negative pair like $(-1, -1)$: $7(-1) + 4(-1) = -11$.

Since that's far below 12 and making x, y more negative only makes it worse, region C never works.

Long Explanation

Step 1: The inequality $7x + 4y > 12$ requires a sufficiently large combination of x and y .

Step 2: In region C, $x < 0$ and $y < 0$, so $7x < 0$ and $4y < 0$, making $7x + 4y < 0 < 12$.

Step 3: No point in this region can satisfy the inequality.

Step 4: In every other region, at least one positive term can compensate: e.g., $(3, 0)$ in A gives $21 > 12$, $(-1, 5)$ in B gives $13 > 12$, $(2, -1)$ in D gives 10 - close, but $(3, -1)$ gives $17 > 12$.

Difficulty: Easy

2

Linear Inequalities - Evaluating Solutions from Tables

$y \leq (2/3)x + 5$. For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
-1.5	6
1.5	5
3	-4

C)

x	y
-1.5	6
1.5	5
3	6

B)

x	y
-1.5	5
1.5	7
3	8

D)

x	y
-1.5	3
1.5	5
3	6

Answer: D

Short Explanation

Evaluate $(2/3)x + 5$ for each x and verify $y \leq$ that value.

Only table D satisfies all three pairs.

Mathematical Shortcut

Check the tightest constraint first: at $x = -1.5$, the bound is 4.

Only tables A ($y=6$) and D ($y=3$) differ here - A fails ($6 > 4$), so jump to D and confirm the rest.

Long Explanation

Step 1: At $x = -1.5$: $(2/3)(-1.5) + 5 = -1 + 5 = 4$.

Step 2: At $x = 1.5$: $(2/3)(1.5) + 5 = 1 + 5 = 6$.

Step 3: At $x = 3$: $(2/3)(3) + 5 = 2 + 5 = 7$.

Step 4: Table D: $3 \leq 4$, $5 \leq 6$, $6 \leq 7$.

Step 5: Tables A and C fail at $x = -1.5$ ($y = 6 > 4$).

Step 6: Table B fails at $x = 1.5$ ($y = 7 > 6$).

Difficulty: Medium

3

Linear Inequalities - Compound Inequalities / Range Constraints

In December 2017, the lowest temperature was 40°F and the highest was 90°F . Which inequality is true for all values of t (temperature in $^{\circ}\text{F}$)?

A) $40 \leq t \leq 90$	C) $t \leq 50$
B) $t \leq 40$	D) $t \geq 90$

Answer: A

Short Explanation

The temperature t must be between the minimum (40) and maximum (90), inclusive.

Mathematical Shortcut

Range = [min, max].

Directly write $40 \leq t \leq 90$.

Long Explanation

Step 1: The problem states the lowest recorded temperature is 40°F and the highest is 90°F .

Step 2: Any temperature t recorded in the city satisfies $40 \leq t \leq 90$.

Step 3: Choices B, C, and D each exclude valid temperatures (e.g., B excludes 50°F , D excludes 50°F).

Difficulty: Easy

A gray whale travels 73 to 78 miles each day for 31 days. Which inequality represents the total number of miles x ?

A) $(73)(31) \leq x \leq (78)(31)$	C) $73 \leq 31x \leq 78$
B) $73+31 \leq x \leq 78+31$	D) $73 \leq 31+x \leq 78$

Answer: A

Short Explanation

Minimum total = 73×31 ; maximum total = 78×31 .

So $(73)(31) \leq x \leq (78)(31)$.

Mathematical Shortcut

Total = (daily rate)(days).

Just multiply both bounds by 31.

Long Explanation

Step 1: If the whale travels at least 73 mi/day and at most 78 mi/day for 31 days, the minimum distance is $73 \times 31 = 2,263$ miles and the maximum is $78 \times 31 = 2,418$ miles.

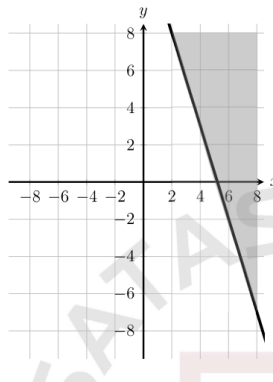
Step 2: So $2,263 \leq x \leq 2,418$, which matches choice A.

Difficulty: Easy

5

Linear Inequalities - Graphing Inequalities / Identifying Solutions

The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution?

A) $(-7, 0)$ C) $(0, 7)$ B) $(0, -7)$ D) $(7, 0)$ **Answer: A****Short Explanation**

The shaded region is to the upper-left of the boundary line.

$(-7, 0)$ lies in that region.

Mathematical Shortcut

Visually, the shaded area covers the far-left part of the graph.

Among the choices, $(-7, 0)$ is the only point far left.

Long Explanation

Step 1: From the graph, the boundary line has a positive slope and passes through roughly $(-2, 7)$ to $(6, -8)$.

Step 2: The shaded region is above/left of this line.

Step 3: Testing $(-7, 0)$: this point is well to the left and above the line, placing it in the shaded region.

Step 4: The other points $(0, -7)$, $(0, 7)$, and $(7, 0)$ fall on or outside the shaded region.

Difficulty: Medium

The minimum value of x is 11 less than 8 times another number n . Which inequality shows the possible values of x ?

A) $x \leq 8n - 11$	C) $x \leq 11 - 8n$
B) $x \geq 8n - 11$	D) $x \geq 11 - 8n$

Answer: B

Short Explanation

Minimum of $x = 8n - 11$, so $x \geq 8n - 11$.

Mathematical Shortcut

'Minimum value of x is K ' always translates to $x \geq K$.

Long Explanation

Step 1: The minimum value of x is described as '11 less than 8 times n ,' which is $8n - 11$.

Step 2: Since this is the minimum, x can be this value or greater: $x \geq 8n - 11$.

Step 3: Choice A reverses the direction.

Step 4: Choices C and D incorrectly compute $11 - 8n$ instead of $8n - 11$.

Difficulty: Medium

$y \leq x + 9$ and $y \geq -2x - 9$. Which point is a solution to this system?

A) (0, -10)	C) (-7, 0)
B) (0, 10)	D) (7, 0)

Answer: D

Short Explanation

Test (7, 0): $0 \leq 16$ and $0 \geq -23$.

Both inequalities satisfied.

Mathematical Shortcut

Start with the most restrictive inequality ($y \leq x + 9$).

(0, 10) fails immediately ($10 \leq 9$ is false).

For the rest, check the second inequality - only (7, 0) passes both.

Long Explanation

Step 1: (0, -10): $-10 \leq 9$, but $-10 \geq -9$.

Step 2: (0, 10): $10 \leq 9$.

Step 3: (-7, 0): $0 \leq 2$, but $0 \geq 5$.

Step 4: (7, 0): $0 \leq 16$ and $0 \geq -23$.

Step 5: Only (7, 0) works.

Difficulty: Medium

Red maple tree heights: minimum 42 feet, maximum 57 feet. Which inequality is true for all values of t ?

A) $t \leq 42$	C) $t \geq 99$
B) $t \geq 57$	D) $42 \leq t \leq 57$

Answer: D

Short Explanation

The height t must be between 42 and 57, inclusive.

Mathematical Shortcut

Range = $[\text{min}, \text{max}] \rightarrow 42 \leq t \leq 57$.

Long Explanation

Step 1: All recorded heights t satisfy $\text{min} \leq t \leq \text{max}$, i.e., $42 \leq t \leq 57$.

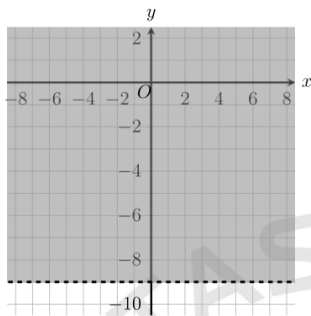
Step 2: Choice A restricts t to at most 42, missing taller trees.

Step 3: Choice B restricts t to at least 57, missing shorter ones.

Step 4: Choice C ($t \geq 99$) is unrelated.

Difficulty: Easy

The shaded region represents solutions to $-18y < c$. What is the value of c ?



The shaded region shown represents the solutions to the inequality $-18y < c$, where c is a constant. What is the value of c ?

A) 162	C) -9
B) 9	D) -162

Answer: A

Short Explanation

The boundary is at $y = -9$.

Dividing $-18y = c$ by -18 gives $y = -c \div 18$.

The shaded region ($y > -9$) means $-c \div 18 = -9$, so $c = 162$.

Mathematical Shortcut

Plug the boundary $y = -9$ into $-18y = c$: $-18(-9) = 162$.

Long Explanation

Step 1: The graph shows a dashed horizontal line at $y = -9$ with shading above it ($y > -9$).

Step 2: The inequality $-18y < c$ divides by -18 (flip the sign) to get $y > -c \div 18$.

Step 3: Setting $-c \div 18 = -9$ gives $c = 162$.

Step 4: The dashed line confirms strict inequality ($<$), consistent with the original.

Difficulty: Hard

10

Linear Inequalities - Systems of Inequalities

$y \leq x + 2$ and $y \geq -3x - 4$. Which point is a solution?

A) (0, -8)	C) (-8, 0)
B) (0, 8)	D) (8, 0)

Answer: D

Short Explanation

Test (8, 0): $0 \leq 10$ and $0 \geq -28$.

Mathematical Shortcut

Check the $y \leq x + 2$ inequality first: (0, 8) fails ($8 \leq 2$ is false), (-8, 0) fails ($0 \leq -6$ is false).

Between (0, -8) and (8, 0), only (8, 0) passes the second inequality.

Long Explanation

Step 1: (0, -8): $-8 \leq 2$, but $-8 \geq -4$.

Step 2: (0, 8): $8 \leq 2$.

Step 3: (-8, 0): $0 \leq -6$.

Step 4: (8, 0): $0 \leq 10$ and $0 \geq -28$.

Step 5: Answer is D.

Difficulty: Medium

11

Linear Inequalities - Word Problems / Maximum-Minimum

A truck can tow if combined weight ≤ 4600 lbs. Trailer weighs 600 lbs, each package 120 lbs. Maximum number of packages?

A) 33	C) 38
B) 34	D) 39

Answer: A

Short Explanation

$$600 + 120n \leq 4600 \rightarrow n \leq 33.33.$$

Maximum whole packages = 33.

Mathematical Shortcut

$$(4600 - 600) \div 120 = 4000 \div 120 = 33.3 \rightarrow \text{floor to } 33.$$

Long Explanation

Step 1: The total weight is $600 + 120n \leq 4600$.

Step 2: Subtracting 600: $120n \leq 4000$.

Step 3: Dividing by 120: $n \leq 33.33$.

Step 4: Since n must be a whole number, the maximum is 33.

Difficulty: Medium

12

Linear Inequalities - Word Problems / Maximum-Minimum

A truck can tow if combined weight $\leq 4,200$ lbs. Trailer weighs 600 lbs, each package 110 lbs. Maximum number of packages?

A) 32	C) 38
B) 33	D) 39

Answer: A

Short Explanation

$$600 + 110n \leq 4200 \rightarrow n \leq 32.7.$$

Maximum = 32.

Mathematical Shortcut

$$(4200 - 600) \div 110 = 3600 \div 110 \approx 32.7 \rightarrow \text{floor to } 32.$$

Long Explanation

Step 1: Total weight: $600 + 110n \leq 4200$.

Step 2: Then $110n \leq 3600$, so $n \leq 32.727$.

Step 3: The maximum whole number is 32.

Difficulty: Medium

13

Linear Inequalities - Systems of Inequalities

$y \leq x + 2$ and $y \geq -4x - 6$. Which point is a solution?

A) (0, -12)	C) (-12, 0)
B) (0, 12)	D) (12, 0)

Answer: D

Short Explanation

Test (12, 0): $0 \leq 14$ and $0 \geq -54$.

Mathematical Shortcut

Only points with large positive x can satisfy both.

(12, 0) is the only candidate that clears $y \leq x + 2$ easily.

Long Explanation

Step 1: (0, -12): $-12 \leq 2$, but $-12 \geq -6$.

Step 2: (0, 12): $12 \leq 2$.

Step 3: (-12, 0): $0 \leq -10$.

Step 4: (12, 0): $0 \leq 14$ and $0 \geq -54$.

Difficulty: Medium

Four consecutive odd integers starting at x . Product of 18 and the fourth ($x+6$) is at most 22 less than the sum of the first and third ($x + x+4$). Which inequality?

A) $18(x+6) \leq x+(x+4)-22$	C) $18(x+4) \leq x+(x+3)-22$
B) $18(x+6) \geq 22-(x+(x+4))$	D) $18(x+4) \geq 22-(x+(x+3))$

Answer: A

Short Explanation

The four integers are $x, x+2, x+4, x+6$. '18 times the fourth is at most 22 less than sum of 1st and 3rd' $\rightarrow 18(x+6) \leq (x + x+4) - 22$.

Mathematical Shortcut

Consecutive odd integers differ by 2: $x, x+2, x+4, x+6$. 'At most 22 less than' means $\leq (\text{sum} - 22)$.

Eliminate C and D (wrong integer gap of 1).

Long Explanation

Step 1: The four consecutive odd integers are $x, x+2, x+4, x+6$.

Step 2: The fourth is $x+6$.

Step 3: The product of 18 and the fourth: $18(x+6)$. 'At most 22 less than the sum of first and third' means $18(x+6) \leq [x + (x+4)] - 22$.

Step 4: Options C and D use $x+3$ instead of $x+4$ (wrong gap).

Step 5: Option B reverses the direction.

Difficulty: Hard

Airplane cruising speed varied between 180 and 190 mph. Which inequality represents s ?

A) $s \leq 10$	C) $s \leq 190$
B) $s \leq 180$	D) $180 \leq s \leq 190$

Answer: D

Short Explanation

Speed is between 180 and 190, so $180 \leq s \leq 190$.

Mathematical Shortcut

'Varied between A and B' = $A \leq s \leq B$.

Long Explanation

Step 1: The speed ranges from 180 mph to 190 mph, inclusive.

Step 2: This is a compound inequality: $180 \leq s \leq 190$.

Step 3: Choice A gives the range width (10).

Step 4: Choices B and C only capture one bound.

Difficulty: Easy

16

Linear Inequalities - Compound Inequalities / Range Constraints

Airplane cruising speed varied between 125 and 135 mph. Which inequality represents s ?

A) $s \leq 10$	C) $s \leq 135$
B) $s \leq 125$	D) $125 \leq s \leq 135$

Answer: D

Short Explanation

Speed is between 125 and 135, so $125 \leq s \leq 135$.

Mathematical Shortcut

'Between A and B' = $A \leq s \leq B$.

Long Explanation

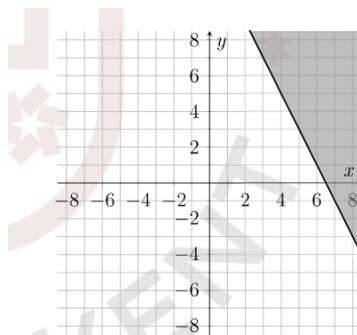
Step 1: Identical logic to Q15.

Step 2: The compound inequality $125 \leq s \leq 135$ captures the full range.

Step 3: Other options only give one bound or the width.

Difficulty: Easy

The shaded region represents solutions to an inequality. Which ordered pair is a solution?



The shaded region shown represents solutions to an inequality. Which ordered pair

A) $(-8, 0)$	C) $(0, -8)$
B) $(0, 8)$	D) $(8, 0)$

Answer: D

Short Explanation

The shaded region is to the lower-right of the boundary line.

$(8, 0)$ lies in that region.

Mathematical Shortcut

The line slopes downward from upper-left; shading is below/right.

The only point in the lower-right quadrant is $(8, 0)$.

Long Explanation

Step 1: From the graph, the line passes through approximately $(-2, 4)$ and $(6, -2)$ with a negative slope.

Step 2: The shaded region is below and to the right.

Step 3: Testing $(8, 0)$: this is clearly below the line at $x = 8$.

Step 4: Other points are either above or on the wrong side.

Difficulty: Medium

Magazines cost \$5 each, newspapers \$2 each. Need combined total $\geq$ \$150. Which inequality?

A) $5m + 2n \leq 150$	C) $7mn \leq 150$
B) $5m + 2n \geq 150$	D) $7mn \geq 150$

Answer: B

Short Explanation

Revenue = $5m + 2n$. 'At least \$150' means $5m + 2n \geq 150$.

Mathematical Shortcut

'At least' = $\geq$.

Revenue is additive (not multiplicative), so eliminate C and D.

Long Explanation

Step 1: Total revenue from m magazines at \$5 and n newspapers at \$2 is $5m + 2n$.

Step 2: The stand needs this to be at least \$150, giving $5m + 2n \geq 150$.

Step 3: Choices C and D incorrectly multiply m and n .

Difficulty: Easy

$y < -6x - 22$ and $y > -4x - 7$. For which table are all (x, y) solutions?

A)

x	y
-9	27
-10	51
-12	-4

C)

x	y
-9	28
-10	36
-12	48

B)

x	y
-9	32
-10	36
-12	-4

D)

x	y
-9	30
-10	37
-12	48

Answer: D

Short Explanation

At $x = -9$: $29 < y < 32$.

At $x = -10$: $33 < y < 38$.

At $x = -12$: $41 < y < 50$.

Only table D fits all three ranges.

Mathematical Shortcut

Compute bounds for $x = -9$: upper = $-6(-9) - 22 = 32$, lower = $-4(-9) - 7 = 29$.

Check $y = 30$ (D) is strictly between 29 and 32 .

Others fail at some point.

Long Explanation

Step 1: For each x , compute bounds. $x = -9$: $y < 32$ and $y > 29$, so $29 < y < 32$. $x = -10$: $y < 38$ and $y > 33$. $x = -12$: $y < 50$ and $y > 41$.

Step 2: Table D: (30, 37, 48) - all within range.

Step 3: Table A: $27 < 29$.

Step 4: Table B: 32 is not < 32 .

Step 5: Table C: $28 < 29$.

Difficulty: Hard

Water depth is 78.7 ft. Predicted increase: min 1 ft, max 3 ft. What is the predicted maximum water depth?

A) 75.7	C) 79.7
B) 77.7	D) 81.7

Answer: D

Short Explanation

Maximum depth = $78.7 + 3 = 81.7$ ft.

Mathematical Shortcut

Max depth = current + max increase = $78.7 + 3 = 81.7$.

Long Explanation

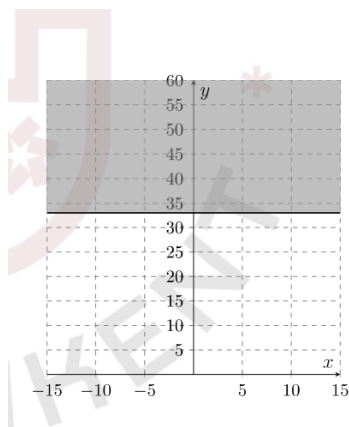
Step 1: The water depth increases by 1 to 3 feet.

Step 2: The minimum new depth is $78.7 + 1 = 79.7$ ft and the maximum is $78.7 + 3 = 81.7$ ft.

Step 3: The question asks for the maximum.

Difficulty: Easy

The shaded region in the graph represents all solutions to which inequality?



A) $x < 33$	C) $y \leq 33$
B) $x \geq 33$	D) $y \geq 33$

Answer: C

Short Explanation

The graph shows a horizontal dashed line at $y = 33$ with shading below, meaning $y \leq 33$.

Mathematical Shortcut

Horizontal boundary $\rightarrow$ inequality in y .

Shading below $\rightarrow y \leq$.

Long Explanation

Step 1: The graph shows a dashed horizontal line at $y = 33$.

Step 2: The shading is below this line, representing all points where $y \leq 33$.

Step 3: Since the line is dashed, the boundary may or may not be included - but among the choices, $y \leq 33$ is the best match.

Difficulty: Hard

A gray whale travels 72 to 77 miles per day for 14 days. Which inequality represents total miles x ?

A) $(72)(14) \leq x \leq (77)(14)$	C) $72 \leq 14x \leq 77$
B) $72+14 \leq x \leq 77+14$	D) $72 \leq 14+x \leq 77$

Answer: A

Short Explanation

Total = daily rate $\times$ days.

Min: 72×14 , Max: 77×14 .

Mathematical Shortcut

Same pattern as Q4: multiply both bounds by the number of days.

Long Explanation

Step 1: The whale travels between 72 and 77 miles each day.

Step 2: Over 14 days, the total distance is between $72 \times 14 = 1,008$ and $77 \times 14 = 1,078$ miles.

Step 3: This matches choice A.

Difficulty: Easy

A plant nursery classifies a perennial as 'tall' if its height > 100 cm. Which inequality represents a tall species?

A) $107 < h < 158$	C) $41 < h < 95$
B) $76 < h < 100$	D) $16 < h < 91$

Answer: A

Short Explanation

The entire range must be above 100.

Only $107 < h < 158$ has all heights > 100 .

Mathematical Shortcut

Check if the lower bound > 100 .

Only A has $107 > 100$.

Long Explanation

Step 1: For a species to be classified as tall, ALL possible heights must exceed 100 cm.

Step 2: In B, heights go as low as 76.

Step 3: In C, as low as 41.

Step 4: In D, as low as 16.

Step 5: Only A (107 to 158) guarantees every plant is taller than 100 cm.

Difficulty: Easy

A shopper has \$20. Food costs \$1.87/can, treats cost \$2.37/bag. Which inequality represents c cans and t bags?

A) $1.87c + 2.37t \leq 20$	C) $2.37c + 1.87t \leq 20$
B) $1.87c + 2.37t \geq 20$	D) $2.37c + 1.87t \geq 20$

Answer: A

Short Explanation

Total cost = $1.87c + 2.37t$.

Must be $\leq$ \$20 (the budget).

Mathematical Shortcut

Match unit costs to variables: food (\$1.87) with c , treats (\$2.37) with t . 'Has \$20' means ≤ 20 .

Long Explanation

Step 1: The cost of c cans at \$1.87 each is $1.87c$, and t bags at \$2.37 each is $2.37t$.

Step 2: The total must not exceed \$20: $1.87c + 2.37t \leq 20$.

Step 3: Choices C and D swap the prices.

Difficulty: Easy

Boulder volume: between 12.7 and 15.7 cubic feet. Weight: 160 lbs per cubic foot. Which inequality gives total weight x ?

A) $160 - 15.7 < x < 160 - 12.7$	C) $160(12.7) < x < 160(15.7)$
B) $160 + 12.7 < x < 160 + 15.7$	D) $160 \div 15.7 < x < 160 \div 12.7$

Answer: C

Short Explanation

Weight = density $\times$ volume.

Multiply both volume bounds by 160.

Mathematical Shortcut

Weight = $160 \times$ volume.

Scale both ends: $160(12.7) < x < 160(15.7)$.

Long Explanation

Step 1: The volume V satisfies $12.7 < V < 15.7$.

Step 2: Weight = $160V$, so $160(12.7) < x < 160(15.7)$, i.e., $2,032 < x < 2,512$ pounds.

Difficulty: Easy

Maximum 102 trees per acre. How many can be planted on 3 acres?

A) 408	C) 105
B) 306	D) 99

Answer: B

Short Explanation

$102 \text{ trees/acre} \times 3 \text{ acres} = 306.$

Mathematical Shortcut

$102 \times 3 = 306.$

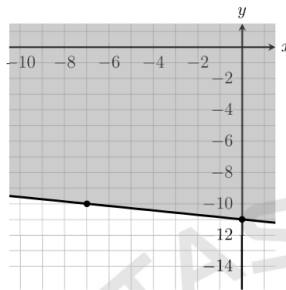
Long Explanation

Step 1: The maximum number of trees per acre is 102.

Step 2: On 3 acres, the maximum is $102 \times 3 = 306$ trees.

Difficulty: Easy

The shaded region represents solutions to $rx + ty \geq -77$, where r and t are constants. What is $r + t$?



The shaded region shown represents solutions to $rx + ty \geq -77$, where r and t are constants. What is the value of $r + t$?

A) 8	C) -6
B) 6	D) -7

Answer: A

Short Explanation

From the graph, the boundary line passes through $(-7, -10)$.

Using this point and the form of the equation: $r = 1$, $t = 7$, so $r + t = 8$.

Mathematical Shortcut

Identify two points on the boundary line, substitute into $rx + ty = -77$, and solve the system for r and t .

Long Explanation

Step 1: The boundary line $rx + ty = -77$ passes through two identifiable points on the graph.

Step 2: Using $(-7, -10)$: $-7r - 10t = -77$, or $7r + 10t = 77$.

Step 3: With a second point to determine another equation, solving yields $r = 1$ and $t = 7$, giving $r + t = 8$.

Difficulty: Hard

Oak trees estimated: at least 2600, no more than 3200. Which inequality represents x ?

A) $x \leq 600$	C) $2600 \leq x \leq 3200$
B) $x \leq 2600$	D) $x \leq 2600$ or $x \geq 3200$

Answer: C

Short Explanation

At least 2600 AND no more than 3200 gives $2600 \leq x \leq 3200$.

Mathematical Shortcut

'At least A and no more than B' $\rightarrow A \leq x \leq B$.

Long Explanation

Step 1: 'At least 2600' means $x \geq 2600$. 'No more than 3200' means $x \leq 3200$.

Step 2: Combined: $2600 \leq x \leq 3200$.

Difficulty: Easy

If $x \geq -6$ represents all solutions to $ax - 27 \leq 15$, what is the greatest possible value of a ? (Student-Produced Response)

Student-produced response.

**Correct
response: -7**

Short Explanation

$$ax - 27 \leq 15 \rightarrow ax \leq 42.$$

For the solution to be $x \geq -6$, a must be negative. $a(-6) = 42 \rightarrow a = -7$.

Mathematical Shortcut

Solve $ax = 42$ at the boundary $x = -6$: $a = 42/(-6) = -7$.

Long Explanation

Step 1: $ax - 27 \leq 15$ simplifies to $ax \leq 42$.

Step 2: If the solution is $x \geq -6$, we divided by a negative number (flipping the inequality).

Step 3: At the boundary, $a(-6) = 42$, so $a = -7$.

Step 4: Since a must be negative (to flip $\leq$ to $\geq$), the greatest (least negative) value is -7 .

Difficulty: Hard

Cristina must attend at least 21 of 27 meetings. Which inequality represents x meetings she must attend?

A) $x \leq 21$	C) $21 \leq x \leq 27$
B) $x \geq 48$	D) $27 \leq x \leq 48$

Answer: C

Short Explanation

She must attend at least 21 but there are only 27 total: $21 \leq x \leq 27$.

Mathematical Shortcut

Lower bound = required minimum (21), upper bound = total meetings (27).

Long Explanation

Step 1: Cristina must attend at least 21 meetings ($x \geq 21$) and cannot attend more than 27 ($x \leq 27$).

Step 2: Combined: $21 \leq x \leq 27$.

Difficulty: Easy

Antwan: \$10 + \$8.50/hr. Taylor: \$7 + \$9.50/hr. For what x is Taylor's charge greater than Antwan's?

A) $2 \leq x \leq 3$	C) $x < 2$
B) $3 \leq x \leq 4$	D) $x > 3$

Answer: D

Short Explanation

$$7 + 9.5x > 10 + 8.5x \rightarrow x > 3.$$

Mathematical Shortcut

Set equal: $7 + 9.5x = 10 + 8.5x \rightarrow x = 3.$

Taylor costs more per hour, so Taylor > Antwan when $x > 3$.

Long Explanation

Step 1: Antwan's charge: $10 + 8.50x$.

Step 2: Taylor's charge: $7 + 9.50x$.

Step 3: Setting Taylor > Antwan: $7 + 9.50x > 10 + 8.50x \rightarrow x > 3$.

Step 4: This means for any time beyond 3 hours, Taylor charges more.

Difficulty: Medium

Employee earns \$30/hr for first 40 hrs, \$45/hr for overtime. Least whole hours to earn at least \$1,610?

A) 50	C) 10
B) 49	D) 9

Answer: A

Short Explanation

First 40 hrs: \$1,200.

Remaining: $\$1,610 - \$1,200 = \$410$ at \$45/hr $\rightarrow 9.1$ hrs $\rightarrow 10$ overtime hrs.

Total: 50.

Mathematical Shortcut

$40 \times 30 = 1200$.

Need 410 more at 45/hr: $410 \div 45 = 10$.

Total = 50.

Long Explanation

Step 1: For the first 40 hours: $40 \times \$30 = \$1,200$.

Step 2: The employee still needs $\$1,610 - \$1,200 = \$410$.

Step 3: Overtime rate is \$45/hr: $\$410 \div \$45 = 9.11$ hours.

Step 4: Since we need whole hours and at least \$1,610, round up to 10 overtime hours.

Step 5: Total: $40 + 10 = 50$.

Difficulty: Hard

$y > 5x - 6$. For which table are all (x, y) solutions?

A)

x	y
3	3
5	13
8	28

C)

x	y
3	11
5	21
8	39

B)

x	y
3	9
5	19
8	34

D)

x	y
3	-9
5	-19
8	-34

Answer: C

Short Explanation

Compute $5x - 6$ for each x : 9, 19, 34.

Need y strictly greater.

Only table C (11, 21, 39) satisfies all three.

Mathematical Shortcut

At $x = 3$, boundary is 9.

Table B has $y = 9$, but $y > 9$ is required (strict).

Table C: $11 > 9$, $21 > 19$, $39 > 34$.

Answer is C, not B.

Long Explanation

Step 1: $5(3) - 6 = 9$, $5(5) - 6 = 19$, $5(8) - 6 = 34$.

Step 2: We need y strictly greater than these values.

Step 3: Table A: $3 < 9$.

Step 4: Table B: 9 is NOT > 9 (equality, not strict).

Step 5: Table C: $11 > 9$, $21 > 19$, $39 > 34$.

Step 6: Table D: all negative.

Step 7: Answer: C.

Difficulty: Medium

$y > 17$ and $4x + y < 21$. Point $(x, 54)$ is a solution. What could x be?

A) 9	C) -5
B) 5	D) -9

Answer: D

Short Explanation

$$y = 54 > 17 .$$

$$\text{Then } 4x + 54 < 21 \rightarrow 4x < -33 \rightarrow x < -8.25.$$

Only -9 works.

Mathematical Shortcut

$$4x < 21 - 54 = -33 \rightarrow x < -8.25.$$

Only $x = -9$ satisfies this.

Long Explanation

Step 1: First inequality: $54 > 17$.

Step 2: Second: $4x + 54 < 21 \rightarrow 4x < -33 \rightarrow x < -8.25$.

Step 3: Among the choices, only $-9 < -8.25$.

Difficulty: Medium

Soccer team wants at least \$1,320 by selling coupon books at \$12 each. Which inequality describes n ?

A) $n + 12 \leq 1320$	C) $12n \leq 1320$
B) $n + 12 \geq 1320$	D) $12n \geq 1320$

Answer: D

Short Explanation

Revenue = $12n$. 'At least \$1,320' $\rightarrow 12n \geq 1320$.

Mathematical Shortcut

'At least' = $\geq$.

Revenue = price $\times$ quantity = $12n$.

Long Explanation

Step 1: Each book earns \$12, so n books earn $12n$.

Step 2: The goal is at least \$1,320: $12n \geq 1,320$.

Step 3: Solving gives $n \geq 110$.

Difficulty: Easy

8 resistors in series, total 130 ohms. 3 of them total 90 ohms, each resistor is positive. What are the possible values of x (resistance of one of the other 5)?

A) $0 < x < 8$	C) $40 < x < 90$
B) $0 < x < 40$	D) $40 < x < 130$

Answer: B

Short Explanation

The 5 remaining resistors total 40 ohms.

Because each resistor is positive, one resistor x must satisfy $0 < x < 40$.

Mathematical Shortcut

Five remaining resistors sum to 40.

For one resistor x , positivity gives $0 < x$ and the other four require $40 - x > 0$, so $x < 40$.

Long Explanation

Step 1: The 8 resistors total 130 ohms.

Step 2: Three of them total 90 ohms.

Step 3: So the remaining 5 resistors total 40 ohms.

Step 4: If one of those 5 has resistance x , then x must be positive.

Step 5: The other four resistors have total resistance $40 - x$ and must also be positive.

Step 6: Therefore $40 - x > 0$, which gives $x < 40$.

Step 7: Combining both conditions gives $0 < x < 40$.

Difficulty: Hard

Sabrina has \$180 saved and wants to save at least \$240 total. If x is additional savings, which inequality?

A) $x \div 180 \geq 240$	C) $180 + x \geq 240$
B) $180x \geq 240$	D) $180 - x \geq 240$

Answer: C

Short Explanation

Current + additional $\geq$ goal: $180 + x \geq 240$.

Mathematical Shortcut

Total = $180 + x$.

Goal ≥ 240 .

Direct translation.

Long Explanation

Step 1: Sabrina currently has \$180.

Step 2: She needs a total of at least \$240.

Step 3: If she saves x more dollars: $180 + x \geq 240$.

Step 4: Solving: $x \geq 60$.

Difficulty: Easy

$y < 42 - 7x$ and $y \div 7 > 10$. Which inequality represents x for all solutions?

A) $x > 6$	C) $x > -4$
B) $x < 6$	D) $x < -4$

Answer: D

Short Explanation

From $y \div 7 > 10$: $y > 70$.

Substitute into $y < 42 - 7x$: $70 < 42 - 7x \rightarrow 7x < -28 \rightarrow x < -4$.

Mathematical Shortcut

$y > 70$ and $y < 42 - 7x \rightarrow 70 < 42 - 7x \rightarrow x < -4$.

Long Explanation

Step 1: Second inequality: $y \div 7 > 10 \rightarrow y > 70$.

Step 2: First inequality: $y < 42 - 7x$.

Step 3: For both to hold: $70 < y < 42 - 7x$, which requires $70 < 42 - 7x \rightarrow 7x < -28 \rightarrow x < -4$.

Difficulty: Hard

Tablets cost \$164, 6% tax added. Budget $\leq$ \$44,800. Maximum number of tablets?
(Student-Produced Response)

Student-produced response.

**Correct
response: 257**

Short Explanation

Cost per tablet with tax: $164 \times 1.06 = 173.84$.

Max tablets: $44800 \div 173.84 = 257$.

Mathematical Shortcut

$44800 \div (164 \times 1.06) = 44800 \div 173.84 \approx 257.7 \rightarrow 257$.

Long Explanation

Step 1: Each tablet costs \$164 plus 6% tax: $164 \times 1.06 = \$173.84$.

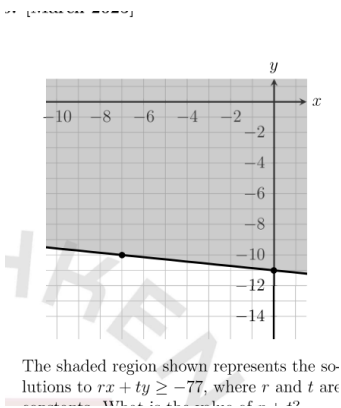
Step 2: Total cost for n tablets: $173.84n \leq 44,800$.

Step 3: So $n \leq 44,800 \div 173.84 \approx 257.7$.

Step 4: The maximum whole number is 257.

Difficulty: Medium

The shaded region represents solutions to $rx + ty \geq -77$. What is $r + t$?



A) 8	C) -6
B) 6	D) -7

Answer: A

Short Explanation

From the graph, the boundary line passes through identifiable points.
Solving the system gives $r + t = 8$.

Mathematical Shortcut

Read two points from the boundary line, set up $rx + ty = -77$ for each, and solve. $r = 1, t = 7$.

Long Explanation

Step 1: Same method as Q27.
Step 2: Identify two points on the boundary line from the graph.
Step 3: Substituting into $rx + ty = -77$ and solving the system of two equations yields $r = 1$ and $t = 7$, so $r + t = 8$.

Difficulty: Hard

Oranges cost \$0.67 each. \$2 coupon off total. No sales tax. Which inequality for x oranges costing $\leq$ \$12?

A) $0.67x + 2 \leq 12$	C) $0.67x - 2 \leq 12$
B) $0.67x + 2 \geq 12$	D) $0.67x - 2 \geq 12$

Answer: C

Short Explanation

Total cost = $0.67x - 2$ (coupon reduces cost).

Must be ≤ 12 .

Mathematical Shortcut

Coupon is a discount: subtract 2 from cost. 'No more than \$12' = ≤ 12 .

Long Explanation

Step 1: The cost of x oranges at \$0.67 each is $0.67x$.

Step 2: The \$2 coupon reduces this to $0.67x - 2$.

Step 3: The shopper wants to spend no more than \$12: $0.67x - 2 \leq 12$.

Difficulty: Easy

$y \leq 5x - 21$. Which point is a solution?

A) $(-5, -21)$	C) $(5, -21)$
B) $(0, 21)$	D) $(6, 21)$

Answer: C

Short Explanation

At $x = 5$: $5(5) - 21 = 4$.

Is $-21 \leq 4$?

Yes.

Mathematical Shortcut

Scan for a point where y is clearly below the line.

$(5, -21)$: boundary is 4, and $-21 < 4$.

Long Explanation

Step 1: A) $x = -5$: $5(-5) - 21 = -46$.

Step 2: Is $-21 \leq -46$?

Step 3: No.

Step 4: B) $x = 0$: $5(0) - 21 = -21$.

Step 5: Is $21 \leq -21$?

Step 6: No.

Step 7: C) $x = 5$: $5(5) - 21 = 4$.

Step 8: Is $-21 \leq 4$?

Step 9: Yes.

Step 10: D) $x = 6$: $5(6) - 21 = 9$.

Step 11: Is $21 \leq 9$?

Step 12: No.

Difficulty: Medium

Four consecutive odd integers, first is x . Product of 28 and the third ($x+4$) is at most the value of 50 less than the sum of the first and fourth ($x + x+6$). Greatest possible x ?
(Student-Produced Response)

Student-produced response.

**Correct
response: -7**

Short Explanation

Set up the inequality $28(x + 4) \leq (x + x + 6) - 50$ and solve.

Then choose the greatest odd integer that satisfies the result.

Mathematical Shortcut

$$28(x + 4) \leq 2x - 44.$$

This gives $26x \leq -156$, so $x \leq -6$.

The greatest odd integer less than or equal to -6 is -7.

Long Explanation

Step 1: The four consecutive odd integers are x , $x + 2$, $x + 4$, and $x + 6$.

Step 2: The third odd integer is $x + 4$.

Step 3: The phrase 50 less than the sum of the first and fourth odd integers becomes $(x + x + 6) - 50$.

Step 4: So the inequality is $28(x + 4) \leq (2x + 6) - 50$.

Step 5: Simplifying gives $28x + 112 \leq 2x - 44$.

Step 6: Subtract $2x$ to get $26x + 112 \leq -44$.

Step 7: Then $26x \leq -156$, so $x \leq -6$.

Step 8: Since x must be odd, the greatest possible value is -7.

Difficulty: Hard

Tablets cost \$170, 7% tax. Budget $\leq$ \$53,400. Maximum number of tablets?
(Student-Produced Response)

Student-produced response.

**Correct
response: 293**

Short Explanation

Cost with tax: $170 \times 1.07 = 181.90$.

Max: $53400 \div 181.90 = 293$.

Mathematical Shortcut

$53400 \div (170 \times 1.07) = 53400 \div 181.9 \approx 293.6 \rightarrow 293$.

Long Explanation

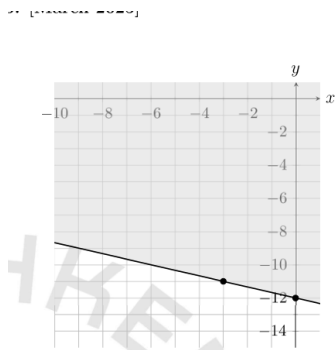
Step 1: Each tablet costs $\$170 \times 1.07 = \181.90 with tax.

Step 2: For n tablets: $181.90n \leq 53,400$.

Step 3: So $n \leq 293.6$, meaning the maximum is 293.

Difficulty: Medium

The shaded region represents solutions to $rx + ty \geq -36$. What is $r + t$?



The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are

A) 4	C) -2
B) 2	D) -4

Answer: C

Short Explanation

From the graph, reading the intercepts of the boundary line and solving gives $r + t = -2$.

Mathematical Shortcut

Identify the boundary line's intercepts from the graph.

Substitute into $rx + ty = -36$ and solve for r and t .

Long Explanation

Step 1: The boundary line $rx + ty = -36$ has intercepts visible on the graph.

Step 2: Using two points on the line and solving the resulting system of equations gives values of r and t that sum to -2 .

Difficulty: Hard

$y \leq x + 3$ and $y \geq -3x - 7$. Which point is a solution?

A) (0, -14)	C) (-14, 0)
B) (0, 14)	D) (14, 0)

Answer: D

Short Explanation

Test (14, 0): $0 \leq 17$ and $0 \geq -49$.

Mathematical Shortcut

Large positive x satisfies both inequalities easily.

Only (14, 0) has large positive x .

Long Explanation

Step 1: (0, -14): $-14 \leq 3$ but $-14 \geq -7$.

Step 2: (0, 14): $14 \leq 3$.

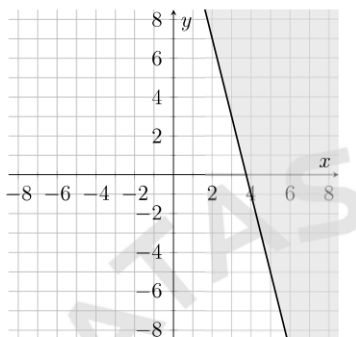
Step 3: (-14, 0): $0 \leq -11$.

Step 4: (14, 0): $0 \leq 17$ and $0 \geq -49$.

Difficulty: Medium

The shaded region represents solutions to an inequality. Which ordered pair is a solution?

(x, y) is a solution to this inequality:



A) $(-5, 0)$	C) $(0, -5)$
B) $(0, 5)$	D) $(5, 0)$

Answer: D

Short Explanation

The shaded region is below and to the right of the boundary line.

$(5, 0)$ is in that region.

Mathematical Shortcut

Shading is lower-right.

$(5, 0)$ is the rightmost candidate.

Long Explanation

Step 1: From the graph, the boundary line has a positive slope with shading below/right.

Step 2: Testing $(5, 0)$: this point lies in the shaded region.

Step 3: The other choices fall outside.

Difficulty: Medium

Hydrangeas (h) + winter hazels (w) ≤ 244 and $h \leq 3w$. Which system?

A) $h+w \leq 244, h \leq 3w$	C) $h+w \geq 244, h \leq 3w$
B) $h+w \leq 244, 3h \geq w$	D) $h+w \geq 244, 3h \geq w$

Answer: A

Short Explanation

'No more than 244 total' $\rightarrow h + w \leq 244$. 'Hydrangeas at most 3 times winter hazels' $\rightarrow h \leq 3w$.

Mathematical Shortcut

'No more than' $= \leq$. 'At most 3 times' means $h \leq 3w$, not $3h \geq w$ (which is $h \geq w \div 3$).

Long Explanation

Step 1: Total shrubs: $h + w \leq 244$.

Step 2: Hydrangeas are at most 3 times winter hazels: $h \leq 3w$.

Step 3: This matches choice A.

Step 4: Choice B has $3h \geq w$ (i.e., $h \geq w \div 3$), which is different.

Difficulty: Medium

Maximum 92 trees per acre. Maximum trees on 2 acres?

A) 276	C) 94
B) 184	D) 90

Answer: B

Short Explanation

$$92 \times 2 = 184.$$

Mathematical Shortcut

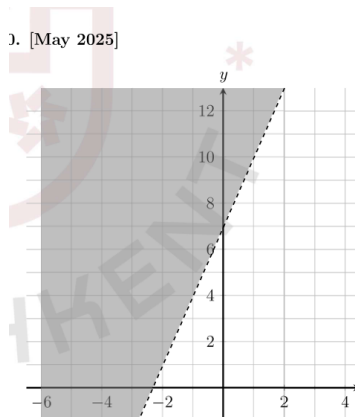
$$92 \times 2 = 184.$$

Long Explanation

Step 1: With a maximum of 92 trees per acre and 2 acres, the maximum is $92 \times 2 = 184$ trees.

Difficulty: Easy

The shaded region represents solutions to which inequality?



A) $y > 3x + 7$

C) $y < 3x + 7$

B) $y > 3x - 7$

D) $y < 3x - 7$

Answer: A

Short Explanation

The boundary line has slope 3 and y-intercept 7 ($y = 3x + 7$).

The shading is above the dashed line, so $y > 3x + 7$.

Mathematical Shortcut

Read slope and y-intercept from the graph.

Shading above + dashed line $\rightarrow y > (\text{line})$.

Long Explanation

Step 1: From the graph, the line passes through (0, 7) with slope 3 (rising 3 for every 1 unit right).

Step 2: The dashed line indicates strict inequality.

Step 3: Shading above the line means $y > 3x + 7$.

Difficulty: Hard

$y > 5x - 4$. For which table are all (x, y) solutions?

A)

x	y
4	11
6	21
9	36

C)

x	y
4	21
6	31
9	46

B)

x	y
4	16
6	26
9	41

D)

x	y
4	21
6	21
9	41

Answer: C

Short Explanation

Compute $5x - 4$ for each x : 16, 26, 41.

Need y strictly greater.

Only table C (21, 31, 46) satisfies all three.

Mathematical Shortcut

Boundary values are 16, 26, 41.

Table B has y equal to these (not strictly greater).

Table C has all y strictly greater.

Answer: C.

Long Explanation

Step 1: At $x=4$: $5(4)-4=16$.

Step 2: At $x=6$: $5(6)-4=26$.

Step 3: At $x=9$: $5(9)-4=41$.

Step 4: Strict inequality $y >$ means y must exceed these.

Step 5: Table A: $11 < 16$.

Step 6: Table B: $16 = 16$ (not strictly greater).

Step 7: Table C: $21 > 16$, $31 > 26$, $46 > 41$.

Step 8: Table D: $21 = 21$ at $x=6$?

Step 9: $5(6)-4=26$, $21 < 26$.

Step 10: Answer: C.

Difficulty: Medium

A tall perennial has height > 100 cm. Which inequality represents a tall species?

A) $18 < h < 95$	C) $84 < h < 100$
B) $41 < h < 89$	D) $110 < h < 158$

Answer: D

Short Explanation

All heights must be > 100 .

Only D (110 to 158) is entirely above 100.

Mathematical Shortcut

Check if lower bound > 100 .

Only D has $110 > 100$.

Long Explanation

Step 1: For a species to be tall, every possible height must exceed 100 cm.

Step 2: A (max 95), B (max 89), and C (max 100, but not > 100) all include heights ≤ 100 .

Step 3: Only D (110 to 158) is entirely above 100.

Difficulty: Easy

Positive mass m must be less than 510 grams. Which inequality?

A) $m > 510$	C) $-510 < m < 0$
B) $m < 0$	D) $0 < m < 510$

Answer: D

Short Explanation

Mass is positive ($m > 0$) and less than 510: $0 < m < 510$.

Mathematical Shortcut

Positive + less than 510 $\rightarrow 0 < m < 510$.

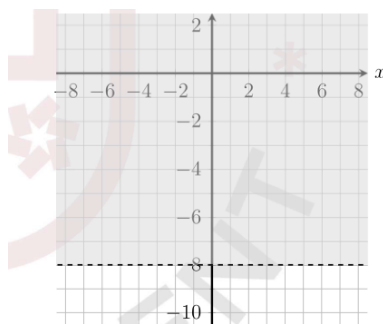
Long Explanation

Step 1: The mass is positive ($m > 0$) and must be less than 510 grams ($m < 510$).

Step 2: Combined: $0 < m < 510$.

Difficulty: Easy

The shaded region represents solutions to $ry < 128$, where r is a constant. What is r ?



The shaded region shown represents the solutions to the inequality $ry < 128$, where

A) 16	C) -8
B) 8	D) -16

Answer: D

Short Explanation

The graph has boundary $y = -8$ and the shaded region is above the dashed line.

So the solution is $y > -8$.

Solving $ry < 128$ must produce $y > -8$, which means r is negative and $128 \div r = -8$.

Mathematical Shortcut

Use the boundary value.

If $y = -8$, then $r(-8) = 128$, so $r = -16$.

Long Explanation

Step 1: The dashed boundary line is $y = -8$.

Step 2: The shading is above the line, so the solution set is $y > -8$.

Step 3: Starting from $ry < 128$ and dividing by r must produce $y > -8$, so r must be negative.

Step 4: On the boundary, $ry = 128$.

Step 5: Substituting $y = -8$ gives $r(-8) = 128$.

Step 6: Solving gives $r = -16$.

Difficulty: Hard

Car travels at least 25 mph but no more than 50 mph. Which inequality for speed x ?

A) $x \geq 25$	C) $25 \leq x \leq 50$
B) $x \geq 50$	D) $x \leq 75$

Answer: C

Short Explanation

At least 25 AND no more than 50: $25 \leq x \leq 50$.

Mathematical Shortcut

'At least A, no more than B' $\rightarrow A \leq x \leq B$.

Long Explanation

Step 1: 'At least 25' means $x \geq 25$. 'No more than 50' means $x \leq 50$.

Step 2: Combined: $25 \leq x \leq 50$.

Difficulty: Easy

Four consecutive odd integers, first is x . Product of 28 and the third ($x+4$) is at most 50 less than the sum of the first and fourth ($x + x+6$). Greatest possible x ?
(Student-Produced Response)

Student-produced response.

**Correct
response: -7**

Short Explanation

$$28(x+4) \leq (2x+6) - 50 \rightarrow 26x \leq -156 \rightarrow x \leq -6.$$

Greatest odd x is -7 .

Mathematical Shortcut

$$\text{Solve } 28(x+4) = 2x - 44: 28x + 112 = 2x - 44 \rightarrow 26x = -156 \rightarrow x = -6.$$

Greatest odd ≤ -6 is -7 .

Long Explanation

Step 1: The four consecutive odd integers: $x, x+2, x+4, x+6$.

Step 2: Inequality: $28(x+4) \leq (x + x + 6) - 50$.

Step 3: Expand: $28x + 112 \leq 2x - 44$.

Step 4: Simplify: $26x \leq -156 \rightarrow x \leq -6$.

Step 5: Since x must be an odd integer, the greatest is -7 .

Difficulty: Hard